PolicyBench

Benchmarking no-tool tax-and-benefit estimation in frontier language models

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Source: Article Notebook

1 Abstract

PolicyBench evaluates whether frontier language models can estimate household tax and benefit outputs from household facts without tools. The benchmark covers the United States and the United Kingdom, uses sampled US Enhanced Current Population Survey (CPS)-derived scenarios and a public UK calibrated transfer dataset, and scores outputs with a bounded 0-100 metric. The manuscript snapshot is a 100-household-per-country public preview, not a protected held-out leaderboard: the current public scenario explorer exposes prompts and reference outputs, so open-set leakage is a central limitation of any public ranking. In the frozen snapshot used for this manuscript (2026-05-15), gpt-5.5 is top-scoring in the US, UK, and shared global views at 70.3, 50.3, and 60.3, respectively. In both countries, multi-step tax quantities and positive-dollar benefit cases are materially harder than zero cases. The live benchmark is available at https://policybench.org.

2 Introduction

Household tax-and-benefit estimation sits between arithmetic and policy discussion. Each case has numeric labels, but generating those labels requires filing status, household composition, income concepts, program thresholds, and jurisdiction-specific rules. PolicyBench measures whether models can map household records to those outputs.

Most current language-model evaluations do not test this mapping directly. General math benchmarks emphasize symbolic manipulation and exact final answers (Cobbe et al. 2021; Hendrycks et al. 2021), while generic question-answering benchmarks emphasize recall or instruction following. PolicyBench instead asks whether models can transform household facts into policy outputs without access to tools or simulators.

This matters for public-facing calculators, analyst workflows, and tax-preparation or screening systems. The benchmark is intended to separate verbal policy fluency from household-level quantitative prediction.

3 Related work

The most relevant prior work is tax and statutory reasoning. SARA frames statutory interpretation, including tax-relevant rule application, as a language-understanding problem (Holzenberger et al. 2021). LegalBench broadens this view and includes tax-oriented numeric tasks such as sara_numeric (Guha et al. 2023). RuleArena evaluates rule-guided reasoning in regulation-style settings (Zhou et al. 2025), and TaxCalcBench evaluates frontier models on tax calculation from structured return-like inputs (Bock et al. 2025). Shanahan et al. report a similar split on the Volunteer Income Tax Assistance (VITA) test: models do better on tax knowledge questions than on open-ended calculation tasks (Shanahan et al. 2025).

PolicyBench also sits within a broader literature on quantitative reasoning benchmarks. Canonical math evaluations such as GSM8K and MATH normalized exact final-answer scoring for numeric tasks (Cobbe et al. 2021; Hendrycks et al. 2021). That norm is less natural for continuous-valued policy outputs, where being within 1 percent is closer than being off by an order of magnitude. Recent work has questioned pure exact-match scoring in numeric QA and temporal reasoning, arguing for error-aware metrics instead (Abbood et al. 2025). PolicyBench follows that direction by combining exactness with near-miss thresholds rather than collapsing everything into exact match alone.

Finance-domain benchmarks provide another relevant precedent. FinQA and TAT-QA both show that realistic quantitative reasoning often requires combining structured numbers with domain knowledge rather than solving stylized school-math problems (Chen et al. 2021; Zhu et al. 2021). PolicyBench differs in that its reference outputs are generated by executable tax-benefit microsimulation rather than annotated document questions, but the underlying motivation is similar: domain-specific quantitative reasoning deserves dedicated evaluation.

Finally, PolicyBench depends on two infrastructure literatures that are not themselves large language model (LLM) benchmarks. One is structured-output reliability, since the benchmark relies on multi-output JSON responses and parse coverage rather than free-form prose (Shorten et al. 2024). The other is tax-benefit microsimulation, where systems such as EUROMOD provide the methodological precedent for evaluating policy rules over household microdata (Sutherland and Figari 2013). There is also operational work on applying and evaluating AI systems in public-benefits settings, including caseworker-assist and Supplemental Nutrition Assistance Program (SNAP)-focused evaluations (Nava Labs 2026, 2025; ZenML LLMOps Database 2025). PolicyBench combines these strands into a public cross-model benchmark over household-level tax and benefit outputs.

4 Benchmark design

PolicyBench asks models to predict all benchmark outputs for a household in a single structured response. One response per household reduces repeated prompt cost and keeps the task at the household-return level rather than turning it into a sequence of unrelated one-output calls.

The benchmark uses a bounded 0-100 score. For amount variables, the score averages four hit rates:

1. exact
2. within 1%
3. within 5%
4. within 10%

For binary outputs, the score is exact accuracy. This keeps 100 as the ceiling while still giving partial credit for near misses on amount outputs. PolicyBench also tracks mean absolute error and related error metrics, but those are secondary to the bounded score. This choice preserves exact-match comparability while avoiding the failure mode that recent numeric-evaluation papers have criticized (Abbood et al. 2025).

Because the four thresholds are nested (exact $\subseteq$ within-1% $\subseteq$ within-5% $\subseteq$ within-10%), averaging their indicator functions is equivalent to assigning step credit by error band: predictions that match exactly receive 1.00, predictions inside 1% but not exact receive 0.75, predictions inside 5% but outside 1% receive 0.50, predictions inside 10% but outside 5% receive 0.25, and predictions outside 10% receive 0.00. The score is therefore a partial-credit rule that rewards tighter agreement more than looser agreement, not an unweighted aggregate of independent hit rates.

All requested US outputs are annual amounts or annual eligibility indicators for tax year 2026; UK outputs are annual amounts or annual eligibility indicators for fiscal year 2026-27. For currency amounts, the "exact" hit rate means within 1 currency unit of the reference value after numeric parsing. Percentage-threshold hit rates use relative error when the reference value is nonzero and the same 1 currency-unit absolute tolerance when the reference value is zero. Binary outputs are parsed as 0 or 1 eligibility flags and scored by exact classification accuracy. Missing or unparseable numeric answers receive zero score for that requested output.

Aggregation proceeds in four steps. First, each household-output prediction receives a 0-100 score. Second, each household receives one model score: requested output rows are weighted inside the household by a blend of equal weighting and absolute PolicyEngine reference-impact share. Person-level coverage flags are scored at the person row, with value-proxy impact weights where available. Third, country scores average those household scores with equal household weight. Fourth, global scores average the US and UK country scores for models run in both countries. The global score is therefore an equal-country, household-equal summary for this benchmark sample, not a universally authoritative measure of tax-benefit competence. Equal-output-group scores and other alternative views are reported as sensitivity checks.

The benchmark requires each model response to include numeric answers and one explanation per requested output. The headline score uses only the numeric answers. Explanations are retained for scenario exploration and qualitative error analysis; they should not be interpreted as faithful traces of model reasoning.

Because explanations are required, the canonical task measures policy estimation under a public-facing structured-response contract, not isolated arithmetic accuracy. Prompt fairness is part of the benchmark contract. The current release uses one prompt template per country, with no model-specific tuning. Models receive the same household facts, the same requested outputs, and no web or tool access. The prompt states that unlisted numeric inputs are 0, unlisted boolean or status facts are false, and household characteristics are constant over the tax-benefit year. Provider-specific differences are limited to the response transport needed to obtain structured output.

4.1 Frozen snapshot and open-set status

Leaderboard positions are version-sensitive, so manuscript claims refer to the frozen source-run exports in Table 1 rather than to the live site. The committed source-run exports are the manuscript artifacts. The public site exposes the current prompts, predictions, explanations, and reference outputs for transparency. This makes the public leaderboard open-set: models, model providers, or benchmark users could learn from the released cases before later runs. Protected leaderboard claims would require a separate held-out or rotating set.

Source: Article Notebook

	Item	Value
0	Snapshot date	2026-05-15
1	Model response date	2026-05-13
2	Policy period	US tax year 2026; UK fiscal year 2026-27
3	Frozen export	paper/snapshot/20260501/runs/us_full_run_20260513_polic
4	Frozen export SHA-256 prefixes	US 30ad2c0bb5cf; UK 7c221d7c1b47
5	Snapshot manifest	paper/snapshot/20260501/manifest.json
6	Response retry artifacts	paper/snapshot/20260501/response_retries
7	Deviation annotations	annotations/full_run_20260513_policyengine_4_4_4_nested
8	US scenarios SHA-256 prefix	b05091225c06
9	US reference outputs SHA-256 prefix	04febaadd091
10	UK scenarios SHA-256 prefix	eee79fb1db36
11	UK reference outputs SHA-256 prefix	6286a238e54a
12	Benchmark spec SHA-256 prefix	532e15ba19bf
13	Frozen prompt payload SHA-256 prefixes	US f3f208702ac6; UK 2f64b11a99aa
14	Scoring code SHA-256 prefix	8f465e64a84b
15	US run label	us_full_run_20260513_policyengine_4_4_4_nested_output
16	UK run label	uk_full_run_20260513_policyengine_4_4_4_nested_output
17	Household sample seed	42
18	PolicyEngine.py	4.4.4

	Item	Value
19	PolicyEngine-US	policyengine-us 1.691.10
20	PolicyEngine-UK	policyengine-uk 2.88.16
21	US data bundle	policyengine-us-data 1.113.1
22	UK runtime bundle	policyengine-uk-data 1.55.5
23	UK scenario source	enhanced_cps_2025.h5
24	UK scenario-source repository	PolicyEngine/policyengine-uk-data
25	UK scenario-source pinned commit	9514dfb7ec607897c9f7122a2e073b922c9fd8b6
26	UK scenario-source SHA-256	199ebc61d29231b4799ad337a95393765b5fb5aede1834b93ff2acec
27	Households	100 US and 100 UK
28	Models	12 shared models
29	Output groups	19 US and 7 UK
30	Condition	No tools, no web access, one structured response per househol
31	Response contract	Numeric answer and non-empty explanation for every requeste

Source: Article Notebook

Table 2: Model run configuration in the fro

	Model	Provider ID	US parsed	UK parsed	Structured
0	gpt-5.5	gpt-5.5	2,188/2,188	700/700	provider s
1	claude-opus-4.7	claude-opus-4-7	2,188/2,188	699/700	provider s
2	grok-4.20	xai/grok-4.20-reasoning	2,188/2,188	700/700	provider s
3	claude-sonnet-4.6	claude-sonnet-4-6	2,104/2,188	665/700	provider s
4	gemini-3.1-pro-preview	gemini/gemini-3.1-pro-preview	2,068/2,188	665/700	provider s
5	gemini-3-flash-preview	gemini/gemini-3-flash-preview	2,167/2,188	694/700	provider s
6	grok-4.3	xai/grok-4.3	2,188/2,188	700/700	provider s
7	gpt-5.4-mini	gpt-5.4-mini	2,188/2,188	700/700	provider s
8	claude-haiku-4.5	claude-haiku-4-5-20251001	2,183/2,188	699/700	provider s
9	gemini-3.1-flash-lite-preview	gemini/gemini-3.1-flash-lite-preview	2,161/2,188	427/700	provider s
10	gpt-5.4-nano	gpt-5.4-nano	2,188/2,188	700/700	provider s
11	grok-4.1-fast	xai/grok-4-1-fast-non-reasoning	1,401/2,188	112/700	provider s

Source: Article Notebook

5 Data and scenario construction

5.1 United States

The US benchmark is built from Enhanced Current Population Survey (CPS)-derived households using PolicyEngine US. The sampled households are filtered to keep a single-tax-unit, single-family, single-Supplemental Poverty Measure (SPM)-unit structure with at least one adult and a supported filing status. The 2024 Enhanced CPS source contains 41,314 households; 30,173 (73.0%) pass the filter and form the eligible draw. The 27.0% excluded by the filter include multi-tax-unit households (e.g., adult roommates), multi-family households, multi-SPM-unit households, and households whose head reports a filing status outside the supported set. These excluded compositions are exactly the kind of cases where federal/state credit allocations and benefit-unit rules become hardest, so the eligible draw is a tractable subset rather than the full distribution of US households. Prompts include nonzero promptable raw inputs across relevant entities rather than a hand-curated summary, so the models see many of the same facts the simulator receives. Filing status is not stated in the prompt; the reference computation infers it from tax-unit role flags. Models therefore see the same household facts that drive the reference filing-status assignment, but they do not receive that assignment as a label.

The current US release evaluates 19 output groups spanning federal income tax, refundable credits, payroll and self-employment tax, state and local income tax, Supplemental Nutrition Assistance Program (SNAP), Supplemental Security Income (SSI), Temporary Assistance for Needy Families (TANF), Affordable Care Act (ACA) premium tax credits, school-meal eligibility, and person-level coverage eligibility for the Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), Medicaid, the Children's Health Insurance Program (CHIP), Medicare, Head Start, and Early Head Start.

The output scope is intentionally narrower than the full PolicyEngine model. Table 3 summarizes the inclusion rule. The benchmark asks for WIC eligibility rather than a WIC dollar amount; WIC dollar values are used only as impact-weight proxies for coverage flags, not as requested model outputs.

Source: Article Notebook

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	Scope decision	Rationale
0	Included	Direct tax, credit, benefit, health-support, and coverage outputs that a housel
1	Excluded	Intermediate tax bases, payroll subcomponents, and outputs that mainly requ
2	ACA Premium Tax Credit	Retained as a deliberate health-support output; when local benchmark premi
3	Binary coverage outputs	Requested as 0/1 eligibility flags and scored as classification tasks; their dolla
4	WIC	The benchmark asks for person-level WIC eligibility. It does not ask models t

Source: Article Notebook

5.2 United Kingdom

The UK benchmark is built from a calibrated public transfer dataset scored through PolicyEngine UK. The current public build starts from a public export of benchmark-compatible households from PolicyEngine US Enhanced CPS, maps those records into UK-facing inputs, and recalibrates them to selected UK targets. The resulting enhanced_cps_2025.h5 artifact is checked in to the public PolicyEngine/policyengine-uk-data GitHub repository; the manuscript pins commit 9514dfb7ec607897c9f7122a2e073b922c9fd8b6 so that a third party can retrieve the exact file used here. The artifact contains 28,532 households; 28,502 (99.9%) pass the eligibility filter that retains households with one benefit unit and one or two adults. This creates a public UK-policy transfer benchmark without publishing restricted household microdata; it is not a representative evaluation over native UK household records and should not be treated as a substitute for Family Resources Survey (FRS)-based UK microdata. The current UK release evaluates seven outputs: Income Tax, National Insurance, Capital Gains Tax, Child Benefit, Universal Credit, Pension Credit, and Personal Independence Payment (PIP). Outputs that depend on status or award facts use prompt-visible facts rather than hidden take-up labels; for example, PIP-positive cases list the daily living and mobility award components used by PolicyEngine.

The UK data path is more synthetic than the enhanced FRS pipeline and inherits limitations from cross-country transfer, calibration choices, and the subset of variables that can be made prompt-visible. It supports the current public cross-country benchmark, but it is not equivalent to an enhanced-FRS-based benchmark and should not be used to make population-representative claims about UK households (Sutherland and Figari 2013).

5.3 Reference-output credibility

PolicyBench treats PolicyEngine outputs as benchmark reference outputs, not as administrative records. The reference source is nevertheless stronger than an ad hoc answer key: PolicyEngine is open source, used for household calculators and reform analysis, and externally checked in specific domains. In the UK, No. 10's data science team adapted PolicyEngine's open-source microsimulation model for experimental policy simulation, with validation against external projections before use (Woodruff 2026; Ghenis 2026). In the US, PolicyEngine reports matching the National Bureau of Economic Research (NBER) TAXSIM-35 model to the cent on the vast majority of cases for the 2021 tax year across hundreds of thousands of tax units per state, with state-specific differences documented in PolicyEngine's integration tests (PolicyEngine 2024; Feenberg and Coutts 1993). We do not restate that comparison as a single percentage because the published source uses qualitative phrasing rather than a headline accuracy number. PolicyEngine has also signed a memorandum of understanding with the Federal Reserve Bank of Atlanta for future validation work against the Atlanta Fed's Policy Rules Database (Ghenis and Makarchuk 2025; Federal Reserve Bank of Atlanta 2026). The Atlanta Fed sources are a caveat rather than evidence of completed validation for this benchmark: they document planned collaboration and the comparison source, not finished checks of the frozen PolicyBench outputs. Taken together, these sources support using PolicyEngine as a transparent reference implementation with partial external validation, but they do not validate every benchmark output.

This does not make PolicyEngine infallible. During benchmark development, we performed a manual, developer-led discrepancy review, with LLM assistance used to triage and summarize candidate cases surfaced by model explanations. Table 4 summarizes the main discrepancy classes reviewed before the frozen snapshot. In those reviewed classes, discrepancies reflected model mistakes, prompt ambiguity later corrected, or upstream data/model issues fixed before the frozen snapshot; the reviewed discrepancy classes did not identify unresolved PolicyEngine reference-output defects.

After freezing the snapshot, we also annotated every wrong model-output row and every scenario-output case with at least one wrong row. Table 5 reports that all 7,664 rows receiving less than full score have row-level and case-level annotations. The final row-level sources are llm_error and parse_contract_failure; no frozen-snapshot wrong row remains classified as a prompt ambiguity, unresolved reference-model issue, unresolved reference-data issue, or needs-review item. In this paper, "LLM response error" therefore includes both parsed numeric/policy errors and response-contract failures. The audit is still developer-led rather than independent external validation, but it is exhaustive over scored misses in the frozen snapshot.

Source: Article Notebook

Table 4: Development discrepancy revi

	Discrepancy class	Review outcome
0	US federal credits	Reviewed cases where explanations omitted EITC or refundable CTC;
1	US SNAP, SSI, and Medicaid	Reviewed false-zero and false-positive explanations; common model fai
2	US payroll and overtime	Reviewed overtime-related cases during development; upstream data/m
3	UK Child Benefit	Reviewed HICBC-related misses; the benchmark now asks for gross Ch
4	UK National Insurance	Reviewed cases where models subtracted pension contributions or used
5	UK UC, Pension Credit, and PIP	Reviewed positive-case misses and false positives; explanations usually

Source: Article Notebook

Table 5: Exhaustive annotation coverage for scored misses in the

	Country	Wrong rows audited	LLM numeric/policy errors	Parse-contract failures	Unresolved prompt
0	US	4533	3489	1044	0
1	UK	3131	2192	939	0

Table 5: Exhaustive annotation coverage for scored misses in the

	Country	Wrong rows audited	LLM numeric/policy errors	Parse-contract failures	Unresolved prompt
2	Total	7664	5681	1983	0

Source: Article Notebook

6 Results

6.1 United States leaderboard

The US leaderboard for the frozen manuscript snapshot is shown in Table 6. The top three models in that snapshot are gpt-5.5 (70.3), claude-opus-4.7 (67.5), and grok-4.20 (65.3).

Source: Article Notebook

Table 6: Top US benchmark models in the frozen manuscript snapshot.

	Model	Score	Exact	Within 10%	Parsed	Total
0	gpt-5.5	70.3	86.8	92.5	2188	2188
1	claude-opus-4.7	67.5	85.1	90.3	2188	2188
2	grok-4.20	65.3	86.8	91.9	2188	2188
3	claude-sonnet-4.6	63.8	85.7	88.9	2104	2188
4	gemini-3-flash-preview	62.4	85.1	88.7	2167	2188

Source: Article Notebook

6.2 United Kingdom leaderboard

The UK leaderboard for the frozen manuscript snapshot is shown in Table 7. The top three models in that snapshot are gpt-5.5 (50.3), gemini-3.1-pro-preview (45.2), and grok-4.20 (45.1).

Source: Article Notebook

Table 7: Top UK benchmark models in the frozen manuscript snapshot.

	Model	Score	Exact	Within 10%	Parsed	Total
0	gpt-5.5	50.3	70.7	91.4	700	700

Table 7: Top UK benchmark models in the frozen manuscript snapshot.

	Model	Score	Exact	Within 10%	Parsed	Total
1	gemini-3.1-pro-preview	45.2	67.3	86.1	665	700
2	grok-4.20	45.1	70.3	89.0	700	700
3	claude-opus-4.7	45.1	69.7	88.6	699	700
4	claude-sonnet-4.6	42.0	69.4	86.1	665	700

Source: Article Notebook

6.3 Global shared-model leaderboard

The global leaderboard averages US and UK scores for the models that have been run in both countries. In the frozen manuscript snapshot, the top three are gpt-5.5 (60.3), claude-opus-4.7 (56.3), and grok-4.20 (55.2). The global table is an equal-country summary for this two-country benchmark sample; it should be read alongside the country tables and sensitivity views rather than as a universally authoritative ordering.

Source: Article Notebook

Table 8: Global shared-model leaderboard across US and UK runs.

	Model	Score	Exact	Within 10%	Parsed	Total
0	gpt-5.5	60.3	78.8	92.0	2888	2888
1	claude-opus-4.7	56.3	77.4	89.4	2887	2888
2	grok-4.20	55.2	78.5	90.5	2888	2888
3	claude-sonnet-4.6	52.9	77.6	87.5	2769	2888
4	gemini-3.1-pro-preview	52.3	74.8	86.2	2733	2888
5	gemini-3-flash-preview	51.3	77.5	86.7	2861	2888
6	grok-4.3	49.1	77.2	85.6	2888	2888
7	gpt-5.4-mini	43.5	76.7	80.9	2888	2888
8	claude-haiku-4.5	42.9	76.1	79.8	2882	2888
9	gemini-3.1-flash-lite-preview	41.7	65.0	68.4	2588	2888

Source: Article Notebook

6.4 Simple baselines

Simple baselines help interpret score levels in a zero-heavy benchmark. Table 9 reports an always-zero response and a median-reference-by-output response on the same frozen household

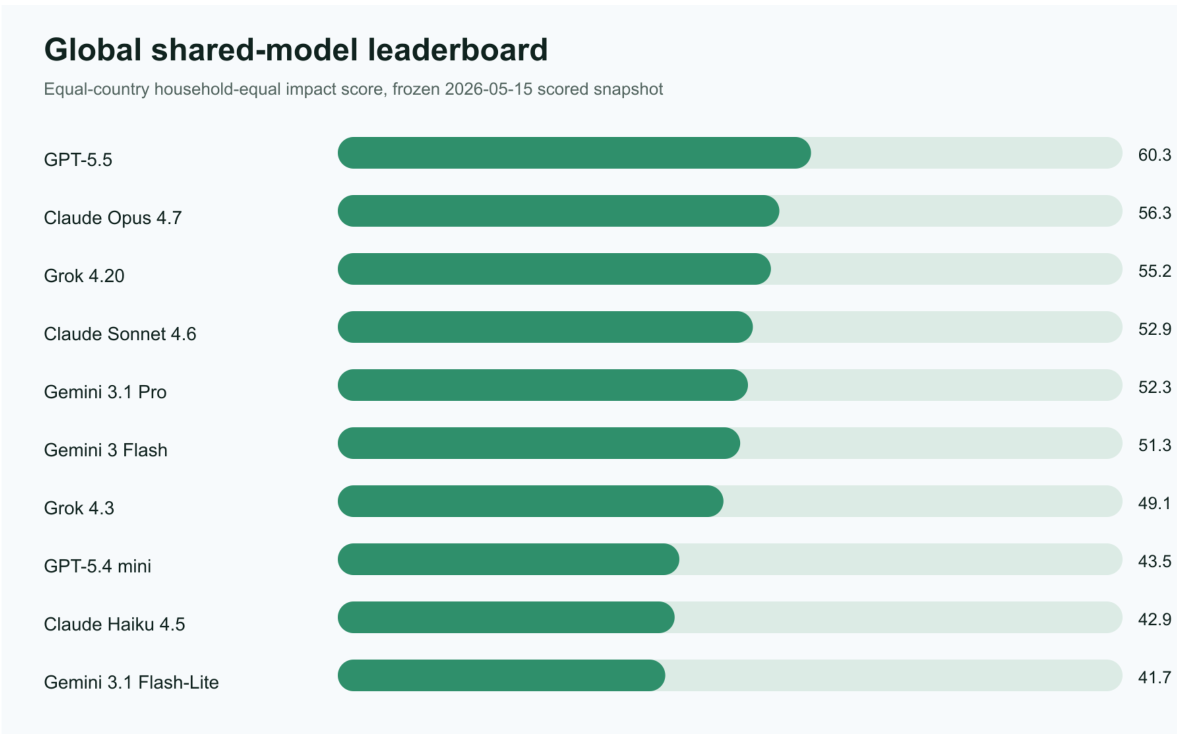

Figure 1: Global shared-model leaderboard in the frozen manuscript snapshot.

sample. These are not model competitors; they show how much of the bounded score can be earned without household-specific policy calculation.

Source: Article Notebook

Table 9: Simple baseline scores on the frozen manuscript snapshot.

	Baseline	Global score	US score	UK score
0	Always zero	26.5	25.3	27.6
1	Median reference by output	24.2	24.6	23.7

Source: Article Notebook

6.5 Sensitivity to benchmark view

The primary leaderboard gives each country equal weight globally and each household equal weight within a country. Table 10 reports several alternative views, including the older equal-output-group score. The top model is stable across several views, but lower ranks move across sensitivity views. The global score is therefore best read as a summary statistic whose interpretation depends on these checks.

Source: Article Notebook

Table 10: Leaderboard sensitivity to alternative scoring views.

	View	Rank 1	Rank 2	Rank 3
0	Main household-equal impact score	gpt-5.5 (60.3)	claude-opus-4.7 (56.3)	grok-4.20 (55.2)
1	Equal-output-group score	gpt-5.5 (85.1)	grok-4.20 (83.5)	claude-opus-4.7 (82
2	Amount outputs only	gpt-5.5 (82.5)	grok-4.20 (80.3)	claude-opus-4.7 (80
3	Positive reference cases only	gpt-5.5 (43.4)	claude-opus-4.7 (40.1)	grok-4.20 (35.9)
4	Zero reference cases only	gpt-5.5 (98.6)	grok-4.20 (98.3)	gemini-3-flash-prev
5	US only	gpt-5.5 (70.3)	claude-opus-4.7 (67.5)	grok-4.20 (65.3)
6	UK only	gpt-5.5 (50.3)	gemini-3.1-pro-preview (45.2)	grok-4.20 (45.1)
7	US binary coverage only	grok-4.20 (97.7)	gemini-3-flash-preview (96.9)	gpt-5.5 (96.8)

Source: Article Notebook

The household-equal impact score weights each requested output row inside a household by a blend of an equal-weight floor and the absolute reference-impact share. The default uses a 0.3 floor. Table 11 reports the top three global ranks under floors of 0.0, 0.1, 0.3, 0.5, and 1.0. Floor 1.0 is equal-row weighting within each household; floor 0.0 is pure impact-share weighting. Rank stability across this range means the main view is not driven by the specific floor choice.

Source: Article Notebook

Table 11: Top global ranks under varying household-equal impact-score floors.

	Floor	Rank 1	Rank 2	Rank 3
0	0.0	gpt-5.5 (49.5)	claude-opus-4.7 (44.7)	grok-4.20 (42.8)
1	0.1	gpt-5.5 (53.1)	claude-opus-4.7 (48.6)	grok-4.20 (46.9)
2	0.3	gpt-5.5 (60.3)	claude-opus-4.7 (56.3)	grok-4.20 (55.2)
3	0.5	gpt-5.5 (67.5)	claude-opus-4.7 (63.9)	grok-4.20 (63.4)
4	1.0	gpt-5.5 (85.6)	grok-4.20 (84.1)	claude-opus-4.7 (83.1)

Source: Article Notebook

6.6 Sampling uncertainty

The manuscript snapshot uses 100 households per country, so small score gaps should not be overinterpreted. Table 12 reports a household-resampling bootstrap over the frozen sample using 1,000 percentile bootstrap draws. It resamples households within each country, recomputes household-equal impact country scores, then recomputes the equal-country global score. The intervals therefore quantify uncertainty from the specific 100-household draw used here. They do not capture (i) prompt-template variance, since the manuscript uses a single template per country with no paraphrase resamples; (ii) decoding stochasticity, since each model's snapshot prediction is a single sample at the provider's default decoding settings; (iii) provider-side drift after 2026-05-13 in alias-resolved model weights; or (iv) reference-output uncertainty inside PolicyEngine. Total uncertainty over those sources is wider than the household-bootstrap intervals.

Source: Article Notebook

Table 12: Household-resampling uncertainty for the global leaderboard.

	Model	Score	95% bootstrap score interval	Bootstrap rank range
0	gpt-5.5	60.3	57.2-63.3	1-1
1	claude-opus-4.7	56.3	53.5-59.1	2-4
2	grok-4.20	55.2	52.2-58.2	2-5
3	claude-sonnet-4.6	52.9	49.9-56.1	3-6
4	gemini-3.1-pro-preview	52.3	48.8-55.7	3-7
5	gemini-3-flash-preview	51.3	48.2-54.5	4-7

Table 12: Household-resampling uncertainty for the global leaderboard.

	Model	Score	95% bootstrap score interval	Bootstrap rank range
6	grok-4.3	49.1	46.2-52.2	5-7
7	gpt-5.4-mini	43.5	40.5-46.7	8-10
8	claude-haiku-4.5	42.9	39.9-46.0	8-10
9	gemini-3.1-flash-lite-preview	41.7	38.4-45.2	8-10

Source: Article Notebook

6.7 Hardest benchmark targets

The lowest-scoring US variables are multi-step tax quantities and sparse positive-dollar benefits. As shown in Table 13, federal income tax before refundable credits, state income tax before refundable credits, payroll tax, and SNAP are the hardest US outputs in the frozen manuscript snapshot.

Source: Article Notebook

Table 13: Hardest US variables by bounded score.

	Variable	Score	Exact	Within 10%
0	federal_income_tax_before_refundable_credits	42.7	35.7	52.1
16	state_income_tax_before_refundable_credits	43.9	39.2	51.8
4	payroll_tax	68.7	56.5	82.2
14	snap	71.9	70.5	73.5
1	federal_refundable_credits	74.3	72.7	76.7

Source: Article Notebook

The same pattern appears in the UK run. Income tax and National Insurance are the two hardest UK outputs, while zero-heavy benefits and capital gains tax score higher overall.

Source: Article Notebook

Table 14: Hardest UK variables by bounded score.

	Variable	Score	Exact	Within 10%
2	income_tax	31.4	19.8	46.8
3	national_insurance	49.2	40.6	63.9
1	child_benefit	69.4	56.9	82.7

Table 14: Hardest UK variables by bounded score.

	Variable	Score	Exact	Within 10%
6	universal_credit	70.6	68.8	73.2
4	pension_credit	82.8	82.8	82.8

Source: Article Notebook

6.8 Zero and positive cases

Overall hit rates can overstate performance on sparse programs because correct zeros are common. Table 15 and Table 16 therefore report within-10% performance for all cases, positive-reference cases, and zero-reference cases. The gap is large for several outputs: models often identify that a program or tax does not apply, but miss the amount when the reference value is positive.

Source: Article Notebook

Table 15: US within-10% performance by zero versus positive reference cases.

	Variable	All cases	Positive cases	Zero cases
0	state_income_tax_before_refundable_credits	54.6	26.7	93.7
1	federal_income_tax_before_refundable_credits	56.2	30.6	94.2
2	snap	76.8	12.9	97.4
3	federal_refundable_credits	80.5	24.6	94.7
4	state_refundable_credits	81.4	4.7	98.9
5	payroll_tax	85.9	83.0	94.9
6	person_medicaid_eligible	89.6	74.3	95.7
7	person_head_start_eligible	92.4	90.9	92.5

Source: Article Notebook

Table 16: UK within-10% performance by zero versus positive reference cases.

	Variable	All cases	Positive cases	Zero cases
0	income_tax	53.4	40.8	92.9
1	national_insurance	71.7	50.7	95.2
2	universal_credit	82.8	24.2	98.0
3	child_benefit	92.6	79.7	100.0
4	pension_credit	92.8	1.5	98.9

Table 16: UK within-10% performance by zero versus positive reference cases.

	Variable	All cases	Positive cases	Zero cases
5	capital_gains_tax	94.5	23.0	99.8
6	pip	99.0	NaN	99.0

Source: Article Notebook

Figure 2: Output-level performance on zero-reference and positive-reference cases.

7 Failure modes

The benchmark surfaces a few recurring failure patterns.

First, models miss positive tax and benefit quantities more often than zero cases. In the US, federal income tax before refundable credits, state income tax before refundable credits, payroll tax, and SNAP are the four lowest-scoring outputs. These require the model to choose the right income concepts, exclusions, program thresholds, and sequencing before applying any final subtraction.

Second, the UK benchmark shows the same split between tax calculations and many benefit outputs. Income tax and National Insurance score below the benefit outputs in the frozen manuscript snapshot. Positive Universal Credit and Pension Credit cases remain difficult, so the result should not be read as a general claim that benefits are easy.

Third, joint accuracy across interacting components is lower than marginal accuracy on either component. Table 17 shows within-10% accuracy for federal_refundable_credits, state_refundable_credits, and the conjunction of both within the same household. The joint hit rate is consistently lower than either marginal hit rate, so leaderboard scores that average across outputs understate how often a model gets a single household's federal/state credit allocation jointly correct.

Source: Article Notebook

Table 17: US within-10% accuracy on federal vs state refundable credits and the household-level joint.

	Model	Federal within 10%	State within 10%	Joint within 10%
8	gpt-5.5	94.0	82.0	77.0
10	grok-4.20	88.0	83.0	76.0
5	gemini-3.1-pro-preview	83.0	79.0	72.0
1	claude-opus-4.7	87.0	78.0	71.0
7	gpt-5.4-nano	79.0	81.0	71.0
2	claude-sonnet-4.6	82.0	80.0	68.0
3	gemini-3-flash-preview	77.0	81.0	66.0
4	gemini-3.1-flash-lite-preview	75.0	80.0	66.0
11	grok-4.3	75.0	81.0	66.0
6	gpt-5.4-mini	73.0	81.0	65.0
0	claude-haiku-4.5	70.0	81.0	62.0
9	grok-4.1-fast	37.0	51.0	31.0

Source: Article Notebook

Fourth, structured-output reliability affects rankings. Missing or unparseable numeric values count as benchmark failures, and Appendix A reports the post-run parser audit. After a conservative reparse of saved raw responses, 271 of 2,400 country-model-household responses still have at least one missing output.

8 Limitations

PolicyBench is not a substitute for a production tax-and-benefit calculator. Several caveats matter:

- model rankings can reflect output-contract reliability as well as policy reasoning
- zero-heavy outputs require separate positive-case interpretation
- the scoring rule is one benchmark view and should be checked against alternative metrics
- the global score is an equal-country, household-equal summary for models run in both countries, not a universal ranking of model quality
- the public UK calibrated transfer dataset is not equivalent to enhanced Family Resources Survey quality and is not population-representative UK microdata
- the public scenario explorer exposes the current test set and reference outputs, so open-set leakage is a prominent limitation rather than a minor implementation detail
- the 100-household manuscript snapshot should be treated as a preview until larger frozen runs are published
- the scored-miss audit is exhaustive over this frozen snapshot, but it is developer-led and not an independent validation of every reference value
- benchmark success should not be interpreted as policy-advice readiness

The current paper is therefore an evaluation of model performance under a specific structured-output benchmark, not a general certification of tax or benefit competence.

9 Conclusion

PolicyBench shows a consistent pattern across both countries. Models often identify non-applicability, but positive tax and benefit amounts remain difficult, especially for multi-step income tax, payroll tax, National Insurance, and positive benefit cases. In the frozen manuscript snapshot, gpt-5.5 is the top-scoring model in both the US and UK.

These results support a narrow conclusion: unaided frontier models still struggle to reproduce selected household-level microsimulation outputs under a structured public benchmark. They do not show that models cannot assist policy analysis, and they do not validate PolicyEngine outputs as administrative truth. They suggest that future evaluations should separate no-tool estimation, tool-using system design, and reference-output validation more explicitly.

Next steps are to expand country coverage, increase frozen sample sizes, improve UK data provenance, and add protected or rotating evaluation sets so that public rankings are less exposed to open-set leakage. The benchmark should also continue reporting sensitivity views, because equal-country household-equal scores are useful summaries only when their weighting choices are visible.

10 Appendix A: Structured-output audit

The benchmark requires one numeric value and one explanation for every requested output. Parse failures are therefore benchmark failures, not missing data to drop from the denominator. After the frozen model run, we audited all rows classified as parse-contract failures, repaired one parser defect, and reran the parser against saved raw responses without making new model calls. Table 18 summarizes that sequence.

Source: Article Notebook

Table 18: Structured-output parser aud

	Step	Finding
0	Initial parse-contract audit	2,025 output rows were classified as parse-contract failures after the frozen mo
1	Parser repair	The parser was extended to accept nested ‘outputs‘ payloads serialized as JSO
2	Recovered outputs	A conservative reparse recovered 42 formerly missing US numeric predictions.
3	Remaining parse failures	1,983 of 34,656 model-output rows remain missing or unparseable after conser
4	Retry unit	Second-chance retries target and replace entire country-model-household resp
5	Preservation rule	The retry artifacts retain the original failed responses, raw retry rows, accept

Source: Article Notebook

The main denominator for response reliability is the country-model-household response: one model answering one household in one country. Table 19 reports that 271 of 2,400 such responses had at least one remaining missing output after the conservative reparse. The output-row count is larger because each response contains multiple requested outputs.

Source: Article Notebook

Table 19: Remaining parse failures by country-model-household response.

	Scope	Responses with any remaining parse failure	Total country-model-household responses
0	Overall	271	2400
1	United States	110	1200
2	United Kingdom	161	1200

Source: Article Notebook

The remaining failures are not one homogeneous bug. Table 20 groups the 1,983 remaining missing output rows, or 5.7% of 34,656 model-output rows, by the raw-response shape observed during the audit. Some rows have no saved raw response and therefore cannot be recovered by parser changes. Others contain malformed JSON, malformed nested outputs strings, or empty tool arguments after repair attempts. The final column counts affected saved-response or chunk groups for that raw-response shape; it is not a mutually exclusive full-response denominator.

Source: Article Notebook

Table 20: Remaining missing output rows by raw-response shape.

	Remaining raw-response shape	Output rows	US rows	UK rows	Affected raw
0	No saved raw response	834	554	280	64
1	Repair attempts returned malformed JSON text	476	168	308	52
2	Tool call with malformed ‘outputs‘ string	291	123	168	31
3	Repair attempts returned malformed ‘outputs‘ string	250	110	140	26
4	Repair attempts returned empty tool arguments	125	89	36	125
5	Invalid saved raw JSON plus top-level ‘outputs‘ object	6	0	6	1
6	Double-nested ‘outputs‘ object	1	0	1	1

Source: Article Notebook

Parse failures are concentrated in a few models. Table 21 reports the model-level response denominator across both countries. Models not shown in the table had no country-model-household response with a remaining parse failure after the conservative reparse.

Source: Article Notebook

Table 21: Remaining response-level parse failures by model.

	Model	Responses with any remaining parse failure	Total responses	Share (%)
0	grok-4.1-fast	121	200	60.5
1	claude-sonnet-4.6	90	200	45.0
2	gemini-3.1-flash-lite-preview	40	200	20.0
3	gemini-3.1-pro-preview	11	200	5.5
4	claude-haiku-4.5	6	200	3.0
5	gemini-3-flash-preview	2	200	1.0
6	claude-opus-4.7	1	200	0.5

Source: Article Notebook

The concentration also differs by country. Table 22 reports the nonzero country-model cells. This is the most direct table for deciding whether a second-chance sensitivity run is worth the cost: a retry should target whole country-model-household responses in these cells and should preserve both the original and retry raw responses.

Source: Article Notebook

Table 22: Remaining response-level parse failures by country and model.

	Country	Model	Responses with any remaining parse failure	Total responses	S
0	UK	grok-4.1-fast	84	100	8
1	UK	gemini-3.1-flash-lite-preview	39	100	3
2	UK	claude-sonnet-4.6	30	100	3
3	UK	gemini-3.1-pro-preview	5	100	5
4	UK	claude-haiku-4.5	1	100	1
5	UK	claude-opus-4.7	1	100	1
6	UK	gemini-3-flash-preview	1	100	1
7	US	claude-sonnet-4.6	60	100	6
8	US	grok-4.1-fast	37	100	3
9	US	gemini-3.1-pro-preview	6	100	6
10	US	claude-haiku-4.5	5	100	5
11	US	gemini-3-flash-preview	1	100	1
12	US	gemini-3.1-flash-lite-preview	1	100	1

Source: Article Notebook

We do not mix replacement values within a single household-model response in the canonical source run. Several outputs in the same response are mechanically related, so replacing only missing variables could combine two inconsistent answer sets. We therefore ran bounded second-chance retries as a sensitivity exercise, accepted a retry only when the full response satisfied the numeric-and-explanation contract, and preserved rejected retry attempts. Round 1 targeted more responses than the numeric-parse denominator above because the retry contract also includes nonempty explanations. Table 23 summarizes the retry rounds.

Source: Article Notebook

Table 23: Full-response retry rounds preserved with the paper snapshot.

	Round	Country	Target responses	Accepted responses	Rejected responses	Estimated cost
0	Round 1	US	138	32	106	$19.03
1	Round 1	UK	184	36	148	$4.94
2	Round 2	US	9	0	9	$0.70
3	Round 2	UK	32	0	32	$0.39

Source: Article Notebook

Accepted retries improved numeric coverage for a few models but did not make repeated contract failures disappear. Table 24 reports model-country cells where accepted retries changed either the household-equal impact score or parsed-output count. Models not shown had no numeric-score or parsed-output change in the accepted-retry sensitivity file. These retry results are preserved as sensitivity artifacts rather than silently replacing the original frozen source-run artifacts.

Source: Article Notebook

Table 24: Score sensitivity after accepted full-response retries.

	Country	Model	Original score	After accepted retries	Score change	Parsed-ou
3	UK	gemini-3.1-flash-lite-preview	23.86	26.71	2.85	77
4	UK	gemini-3.1-pro-preview	45.18	45.68	0.50	14
2	US	gemini-3.1-pro-preview	59.35	62.36	3.01	82
1	US	gemini-3.1-flash-lite-preview	59.59	60.16	0.57	27
0	US	gemini-3-flash-preview	62.40	62.62	0.22	21

Source: Article Notebook

11 Competing interests

The author is affiliated with PolicyEngine, which develops the microsimulation software used to produce the benchmark reference outputs.

Source: Article Notebook

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